

EXPERIMENT NO: 13

EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

AIM:

To write an Embedded C program to add an array of 16-bit numbers and store the results in internal RAM using Keil software.

SOFTWARE REQUIRED:

- Keil

PROGRAM:

```
#include <reg51.h>

void main(void){
    unsigned int i;
    unsigned int array[5] = {
        0x1111,
        0x2222,
        0x8888,
        0x4444,
        0xABCD  };
    unsigned long sum = 0;
    for (i = 0; i < 5; i++)  {
        sum = sum + array[i];  }
    P0 = (unsigned char)(sum & 0xFF);      // Lower 8 bits
    P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
    P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
    while(1); // keep output stable
}
```

The screenshot shows the uVision IDE interface. The main window displays the disassembly of a C program. The left pane shows the 'Registers' window with the 'PC' register at 0x0909. The right pane shows the 'Call Stack - Locals' window, displaying the current function 'MAIN' and its local variables: 'i' (uint), 'array' (array[5] of uint), and 'sum' (ulong). The status bar at the bottom indicates the current address is 0x0909.

Thus the program has been successfully verified and executed.